

Cubic Talbot Revivals and Fixed-Mode Geometry for the Periodic Airy Flow

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Abstract

We give an all-time operator atlas for the periodic Airy equation. Its Fourier flow is a strongly continuous unitary group with least full-space period 2π . At every reduced rational time $2\pi p/q$, the propagator is an exact cubic discrete-Fourier sum of q spatial translations and has global order q . We also classify its full fixed subspace by prime valuations, all minimum periods of finite-spectrum states, and the irrational fixed-space boundary. The receipt independently checks 2,806 strobes and 101 high-precision DFTs.

1 Frozen flow and the global clock

On $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ fix

$$u_t + u_{xxx} = 0, \quad e_n(x) = e^{inx}. \quad (1)$$

Writing $D = -i\partial_x$, Fourier evolution gives

$$U(t)e_n = e^{in^3 t} e_n = e^{itD^3} e_n. \quad (2)$$

The real Fourier multiplier n^3 makes D^3 self-adjoint on its Sobolev domain. Parseval and dominated convergence therefore prove that $U(t)$ is a strongly continuous unitary group on $L^2(\mathbb{T})$.

If $U(t) = I$, the mode $n = 1$ forces $t \in 2\pi\mathbb{Z}$; conversely every integer cube has integral phase at such times. Thus 2π is the least positive return of the *whole* phase space. This is distinct from the often shorter return of an individual state.

2 Complete rational-revival theorem

Theorem 1. *Let $(p, q) = 1$ and $q \geq 1$. At $t = 2\pi p/q$,*

$$U(t)f = \sum_{r=0}^{q-1} A_{p,q}(r) \tau_{2\pi r/q} f, \quad A_{p,q}(r) = \frac{1}{q} \sum_{s=0}^{q-1} e^{2\pi i(ps^3 - sr)/q}, \quad (3)$$

where $\tau_a f(x) = f(x + a)$. Moreover

1. $\sum_r |A_{p,q}(r)|^2 = 1$ and the strobe has exact order q ;
2. its fixed space is

$$\overline{\text{span}}\{e_n : q \mid n^3\} = \overline{\text{span}}\{e_n : L(q) \mid n\}, \quad L(q) = \prod_{\ell^a \parallel q} \ell^{\lceil a/3 \rceil};$$

3. if $t/(2\pi)$ is irrational, the fixed space of $U(t)$ consists exactly of constants.

Proof. The identity

$$(n + q)^3 - n^3 = q(3n^2 + 3nq + q^2)$$

makes the multiplier in (2) q -periodic in n . Finite Fourier inversion yields (3); its unit-modulus input vector gives Parseval. The q th power is the identity, while the mode-one phase forces any order to be divisible by q . Finally, $q \mid n^3$ is equivalent prime by prime to $v_\ell(n) \geq \lceil v_\ell(q)/3 \rceil$. At irrational time, fixing a nonzero mode would make $n^3 t/(2\pi)$ an integer, a contradiction.

3 Individual periods and independent receipt

Let a nonconstant state have finite nonzero Fourier support S . Simultaneous return requires $Tn^3 \in 2\pi\mathbb{Z}$ for every $n \in S$, hence its least positive period is

$$T_S = \frac{2\pi}{\gcd\{|n|^3 : n \in S, n \neq 0\}}. \quad (4)$$

Constants are fixed at every time and have no least positive period. Formula (4) separates state recurrence from the global 2π clock.

The producer covers all reduced (p, q) through $q = 96$: 2,806 exact phase vectors are retained by cryptographic digest, together with the valuation stride and a 1,025-mode fixed-count window. For every reduced pair through $q = 18$, it also stores the full 90-decimal DFT and inverse: 101 transforms.

receipt	rows	independent tests	mathematical sentinel
rational strobes	2,806	50,765 total	mode-one order, $q \mid n^3$
cubic DFT	101	inverse + Parseval	translated-copy identity
finite supports	10	gcd reconstruction	least state period
symbolic/modular	—	301,200	cubic and valuation laws
hostile mutations	41	41 rejected	repaired payload hashes

The checker imports no producer function. It reconstructs modular phases, prime valuations, complex coefficients, inverse DFTs, and support gcds. Finite rows test implementation; Theorem 1 is not inferred from a cutoff.

4 Conclusion and limitations

One theorem now controls the complete periodic Airy group, every rational revival, every sampled fixed subspace, and every finite-support period. It does not classify nonperiodic third-order boundary conditions, evaluate all cubic Gauss sums in closed form, or extend to nonlinear KdV.

References

- [1] B. Pelloni and D. A. Smith, “Revivals, or the Talbot effect, for the Airy equation,” *Studies in Applied Mathematics* (2024), doi:10.1111/sapm.12699.
- [2] G. Farmakis et al., “Beyond periodic revivals for linear dispersive PDEs,” *Proceedings of the Royal Society A* 477 (2021), doi:10.1098/rspa.2021.0241.